

SHUYU GAN

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EDUCATION

University of Minnesota, Twin Cities

Sep. 2025 – Present

Ph.D. Student in Computer Science

Research Areas: Agentic AI, LLM Alignment and Personalization, and Inference-Time Reasoning

Advisors: **Dongyeop Kang** and **Mingyi Hong**

Zhejiang University

Sep. 2020 – Jun. 2024

Bachelor of Engineering with **Honors** in Computer Science, Chu Kochen Honors College

GPA: **3.94/4.00**

RESEARCH INTERESTS

Agentic AI; Multimodal and Computer-Use Agents; LLM Alignment and Personalization; Inference-Time Reasoning

INDUSTRY RESEARCH EXPERIENCE

Applied Sciences Group, Microsoft

Jun. 2026 – Present

Research Intern

Mentor: *Yinheng Li*

- Investigating whether AI agents can autonomously supervise model training.

PUBLICATIONS

Do LLMs Recognize Your Latent Preferences? A Benchmark for Latent Information Discovery in Personalized Interaction *COLM 2026*

Ioannis Tsaknakis, Bingqing Song, **Shuyu Gan**, Dongyeop Kang, Alfredo Garcia, Gaowen Liu, Charles Fleming, Mingyi Hong

Scaling Unverifiable Rewards: A Case Study on Visual Insights *Findings of ACL 2026*

Shuyu Gan, James Mooney, Pan Hao, Renxiang Wang, Mingyi Hong, Qianwen Wang, Dongyeop Kang

A2P-Vis: An Analyzer-to-Presenter Agentic Pipeline for Visual Insights Generation and Reporting *IEEE VIS 2025 VISxGenAI Workshop, Honorable Mention*

Shuyu Gan, Renxiang Wang, James Mooney, Dongyeop Kang

Agent S: An Open Agentic Framework that Uses Computers Like a Human *ICLR 2025*

Saaket Agashe, Jiuzhou Han, **Shuyu Gan**, Jiachen Yang, Ang Li, Xin Eric Wang

Cultural Self-Adaptive Multimodal Gesture Generation Based on Multiple Culture Gesture Dataset *ACM Multimedia 2023*

Jingyu Wu, Shi Chen, **Shuyu Gan**, Weijun Li, Changyuan Eric Yang, Lingyun Sun

RESEARCH EXPERIENCE

Selective Test-Time Scaling for Visual Insight Generation

Jul. 2025 – Present

Graduate Research Assistant, University of Minnesota

Advisor: *Dongyeop Kang*

- Introduced **Selective Test-Time Scaling**, a process-based refinement framework that reallocates inference-time compute and applies stage-local pruning to reduce error accumulation in unverifiable multi-agent tasks.
- Built an end-to-end multi-agent data-analysis pipeline for profiling raw datasets, generating and verifying visual insights, and producing reports.
- Developed an LLM-as-a-judge evaluation framework and identified the judge most aligned with human preferences (*Kendall's $\tau = 0.78$*) through stratified sampling and pairwise preference annotations.
- Improved mean report quality from 74.7 to 78.4 and reduced variance by 25% under a fixed compute budget.

Agent S: A General-Purpose Computer-Use Agent*Research Intern, ERIC Lab, University of California, Santa Cruz*

Jun. 2024 – Dec. 2024

Advisor: Xin Eric Wang

- Co-developed Agent S, an MLLM-based agent for autonomous computer use, combining experience-augmented hierarchical planning, continual memory updates, and an Agent-Computer Interface.
- Built and maintained the codebase; designed the ID-based interface; and integrated online search, tree search, world-model reasoning, and accessibility-tree information for visual grounding.
- Established a new state of the art on OSWorld, improving absolute success rate by 9.37 percentage points (83.6% relative).

Concept-Guided Diffusion Model Training*Research Intern, mlPC Lab, University of California, San Diego*

Jul. 2023 – Sep. 2023

Advisor: Zhuowen Tu

- Incorporated end-to-end conceptual guidance into diffusion-model training by jointly training a generative adversarial network with a Diffusion Transformer (DiT).
- Trained and fine-tuned DiT baselines on CelebA-HQ with perceptual and adversarial guidance, reducing FID from 13.77 to 11.63.

Culture-Aware Co-Speech Gesture Generation*Research Assistant, International Design Laboratory, Zhejiang University*

Feb. 2023 – May 2023

Advisor: Lingyun Sun

- Created and annotated the Multiple Culture Gesture Dataset, comprising 10 cultures and 10,000 motion segments.
- Co-developed a cultural self-adaptive gesture-generation network, implementing its text and audio encoders and cultural feature extractor and introducing adversarial training.
- Improved average FGD from 53.7 to 48.0 and Culture Deception Rate from 33.63% to 39.87%; the work was accepted to ACM Multimedia 2023.

TEACHING EXPERIENCE**Lead Teaching Assistant, CSCI 5541: Natural Language Processing***University of Minnesota*

Fall 2025

AWARDS AND HONORS

Outstanding Graduate of Zhejiang University	2024
Honorary Certificate, Chu Kochen Honors College, Zhejiang University	2024
First-Prize Scholarship and Outstanding Student, Zhejiang University (Top 2%)	2022–2023
Top-Notch Scholarship for Basic Disciplines, Zhejiang University (Top 5%)	2020–2022
Second Prize, National College Mathematics Competition	2021

SKILLS**Programming:** Python, C/C++, Java, Verilog, Assembly**Frameworks and Platforms:** PyTorch; Linux (Ubuntu), Windows**English:** TOEFL iBT 110 (R 29, L 28, S 23, W 30); GRE Verbal 154, Quantitative 170, Analytical Writing 3.5